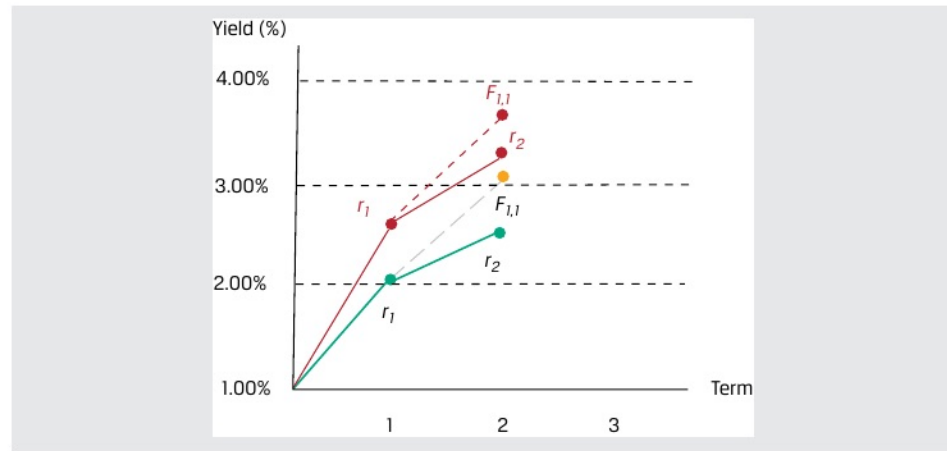




Forward Exchange Rates Using No Arbitrage

We now extend the principle of cash flow additivity to an economic scenario involving different currencies.

Assume that you have USD1,000 to invest for six months. You are considering a riskless investment in either US or Japanese six-month government debt. Let's assume that the current exchange rate between Japanese yen (JPY) and US dollars (USD) is 134.40 (i.e., JPY134.40 = USD1). The six-month Japanese yen risk-free rate is assumed to be 0.05 percent, and the six-month US dollar risk-free rate is 2.00 percent. This example assumes continuous compounding.

Investment strategy 1: Invest USD1,000 in a six-month US Treasury bill

At time $t = 0$: Invest USD1,000 at the 2.00 percent US-dollar risk-free rate for six months.

At time $t = T$ in six months: Receive USD1,010.05 ($= 1,000e^{(0.02 \times 0.5)}$).

Investment strategy 2: Convert the USD1,000 into Japanese yen at the current exchange rate of 134.40, invest in a six-month Japanese Treasury, and convert this known amount back into US dollars in six months at a six-month forward exchange rate set today of 133.096.

At time $t = 0$: Convert USD1,000 into JPY134,400. Lend the JPY134,400 at the 0.05 percent JPY risk-free rate for six months.

At time $t = T$ in six months: Receive JPY loan proceeds of 134,433.60 ($= 134,400e^{(0.0005 \times 0.5)}$). Exchange JPY loan proceeds for US dollars at the forward rate of 133.096 to receive USD1010.05 ($= 134,433.60/133.096$).

These two strategies are economically equivalent in that both involve investing USD1000 at $t = 0$ and receiving USD1010.05 in six months. The element that links these two strategies is the six-month forward exchange rate of 133.096 JPY/USD. If this forward rate is set above or below 133.096, an arbitrage opportunity would exist for investors converting between Japanese yen and US dollars. Exhibit 12 provides a visual layout of the two strategies.